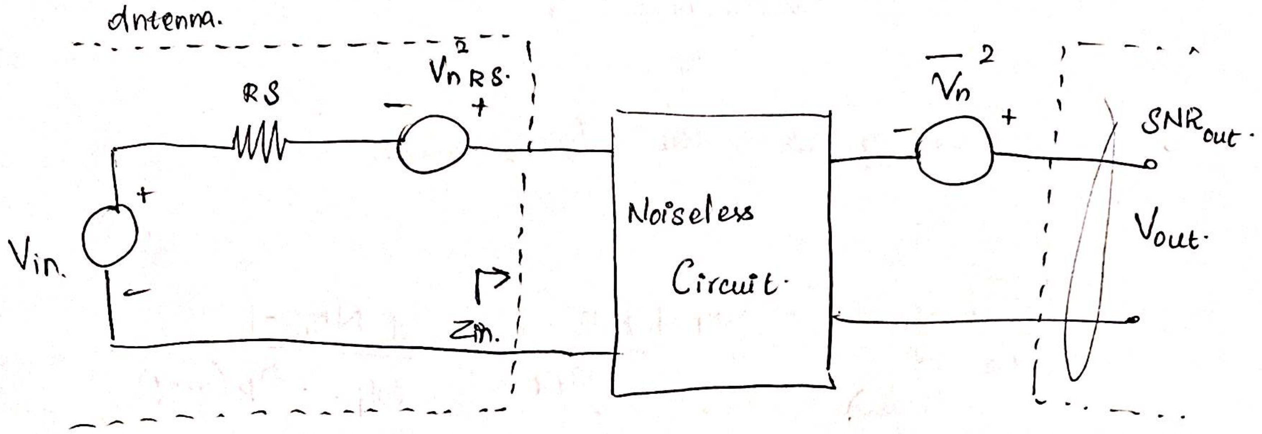
Sheena Selvakumar.

UIN : 537000654.

1, System_NF

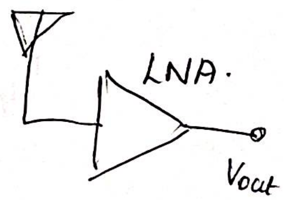
$$\text{Noise figure} = \frac{SNR_{in}}{SNR_{out}}$$

$$\text{Noise figure in dB} = 10 \log \left(\frac{SNR_{in}}{SNR_{output}}\right)$$



For a single stage LNA,

$$SNR_{in} = S_{in}/N_{in} = S_{in} = \underbrace{\left(\frac{Z_{in}}{Z_{in}+R_S}\right)}_{\alpha} V_{in}.$$



$$S_{in} = \alpha V_{in}.$$

$$\boxed{\text{Power} \Rightarrow \alpha^2 V_{in}^2}$$

$$\boxed{SNR_{in} = \frac{\alpha^2 V_{in}^2}{\alpha^2 (V_{nRS})^2}}$$

$$N_{in} = \left[\left(\frac{Z_{in}}{Z_{in}+R_S}\right) V_{nRS}.\right]^2$$

$$= \alpha^2 (V_{nRS})^2.$$

$$\boxed{SNR_{out} = \frac{\left((V_{in}A_V)\alpha\right)^2}{\overline{V_n}^2 + (\alpha V_{RS} A_V)^2}}$$

$$\text{Noise factor} = \frac{SNR_{input}}{SNR_{output}} \Rightarrow \frac{\cancel{\alpha^2 V_{in}^2}}{\alpha^2 (V_{nRS})^2} \times \frac{\overline{V_n}^2 + (\alpha V_{RS} A_V)^2}{(V_{in} A_V \alpha)^2}.$$

$$= 1 + \frac{\overline{V_n}^2}{\alpha^2 A_V^2} \times \frac{1}{V_{nRS}^2}$$

$$F_1 = 1 + \frac{N_1}{N_i G_1}.$$

$$F_2 = 1 + \frac{N_2}{N_i^{\circ} G_2}.$$

$$\text{Nout of second stage} = \left[\underbrace{\overbrace{(N_i^{\circ} G_1) + N_1}^{\text{Noise 1 output amplified by stage 2.}}}_{\text{Noise output of stage 1.}}\right] G_2 + N_2.$$

So According to the Friis formula.

$$NF_{total} \text{ (of all stages)} = NF_1 + \frac{NF_2 - 1}{AP_1} + \dots + \frac{NF_m - 1}{AP_1 \dots Ap(m-1)}$$

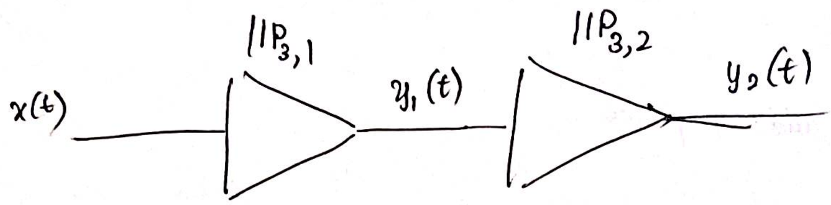
So,

$$F_{sys} = 10^{NF_1/10} + \frac{10^{NF_2/10} - 1}{10^{A_1/10}} + \frac{10^{NF_3/10} - 1}{10^{\frac{A_1+A_2}{10}}} + \dots\dots$$

System Noise factor,

$$\text{System-NF} = 10\log_{10}\left(10^{NF_1/10} + \frac{10^{NF_2/10} - 1}{10^{A_1/10}} + \frac{10^{NF_3/10} - 1}{10^{\frac{A_1+A_2}{10.}}} + \dots\right)$$

Derivation of Cascaded IIP3,



$$y_1(t) = \alpha_1 x(t) + a_2 x^2(t) + a_3 x^3(t).$$

$$y_2(t) = \beta_1 y_1(t) + \beta_2 y_1^2(t) + \beta_3 y_1^3(t).$$

$$y_2(t) = \beta_1 \left[\alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)\right] + \beta_2\left[\alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)\right]^2$$

$$+ \beta_3 \left[\alpha_1 x(t) + \alpha_2 x^2(t) + \alpha_3 x^3(t)\right]^3.$$

$$y_2(t) = \alpha_1\beta_1 x(t) + (\alpha_3\beta_1 + 2\alpha_1\alpha_2\beta_2 + \alpha_1^3\beta_3)x^3(t) + \dots$$

Thus,

$$A_{IP3} = \sqrt{\frac{4}{3}\left|\frac{\alpha_1\beta_1}{\alpha_3\beta_1 + 2\alpha_1\alpha_2\beta_2 + \alpha_1^3\beta_3}\right|}$$

For more stages:

$$\frac{1}{A_{IP3}^2} \simeq \frac{1}{A_{IP3,1}^2} + \frac{\alpha_1^2}{A_{IP3,2}^2} \dots$$

$$P_{IM3} \propto \frac{P_{in}^{3}}{IIP3^{2}}$$

intermodulation products add in power:

$$\frac{1}{IIP3_{sys}} = \frac{1}{IIP3_1} + \frac{G_1}{IIP3_2} + \frac{G_1 G_2}{IIP3_{,3}} + \cdots$$

Refering everything to the input

$$\frac{1}{IIP3_{sys}} = \frac{1}{IIP3_1} + \frac{1}{IIP3_2 / G_1} + \frac{1}{IIP3_3/(G_1 G_2)} + \cdots$$

Convert IIP3 and gains from dB to linear

$$IIP3_i = 10^{\frac{IIP3_i(dBm)}{10}}$$

$$G_i = 10^{(A_i/10)}$$

Substitute into the sum,

$$IIP3_{sys} = \frac{1}{\frac{1}{10^{\left(\frac{IIP3_1}{10}\right)}} + \frac{1}{10^{\left(\frac{IIP3_2 - A_1}{10}\right)}} + \frac{1}{10^{\left(\frac{IIP3_3 - (A_1 + A_2)}{10}\right)}} + \cdots}$$

IIP_3, in dBm,

$$System\_IIP_3 = 10\log_{10}\left(\frac{1}{\frac{1}{10^{(IIP3/10)}} + \frac{1}{10^{(IIP3_2 - A_1)/10}} + \frac{1}{10^{(IIP3_3 - A_1 - A_2)/10}} + \cdots}\right)$$

Matlab implementation:

```
clc;

clear;

%% ====================== INPUTS ======================

% Values in dB

NF   = [3 , 18, 25, 30, 35];      % Noise Figure in dB

gain = [17, 12, 6, 0, 60];      % Gain in dB

IIP  = [-10, 0, 12, 20, 25];    % Input-referred IP3 in dBm

% Resistance value

R = 50;

%% ====================== dB TO VOLTAGE CONVERSION
======================

fprintf('db-v for NF: n');

for i = 1:length(NF)

    db_v= sqrt(8 * R * (10^ (NF(i)/10)) * (1e-3));

    fprintf('NF%d (%.1f dB): %.4f V n', i, NF(i), db_v);

end

fprintf(' ndb-v for gain: n');

for i = 1:length(gain)

    db_v= sqrt(8 * R * (10^ (gain(i)/10)) * (1e-3));

    fprintf('gain%d (%.1f dB): %.4f V n', i, gain(i), db_v);

end

fprintf(' ndb-v for IIP: n');

for i = 1:length(IIP)

    db_v= sqrt(8 * R * (10^ (IIP(i)/10)) * (1e-3));
```

```
    fprintf('IIP%d (%.1f dBm): %.4f V n', i, IIP(i), db_v);
end

%% ====================== SYSTEM NOISE FIGURE
======================

% Convert to linear
F = 10.^(NF/10);     % Noise factor
G = 10.^(gain/10);  % Gain (linear)

% Friis formula
F_sys = F(1);
gain_prod = 1;

for k = 2:length(F)
    gain_prod = gain_prod * G(k-1);
    F_sys = F_sys + (F(k) - 1)/gain_prod;
end

% Convert back to dB
System_NF_dB = 10*log10(F_sys);

fprintf(' n====================== RESULTS ======================
n');
fprintf('System Noise Figure = %.2f dB n', System_NF_dB);

%% ====================== SYSTEM IIP3 ======================

% Convert IIP3 to linear power (mW)
IIP_ln = 10.^(IIP/10);
```

```
% Cascaded IIP3 calculation
inv_IP3_sys = 1/IIP_lin(1);
gain_prod = 1;

for k = 2:length(IIP_lin)
    gain_prod = gain_prod * G(k-1);
    inv_IP3_sys = inv_IP3_sys + gain_prod/IIP_lin(k);
end

IIP3_sys_lin = 1/inv_IP3_sys;

% Convert back to dBm
System_IP3_dBm = 10*log10(IIP3_sys_lin);

fprintf('System IIP3 = %.2f dBm n', System_IP3_dBm);
fprintf('=============================================
n');
```

The output of the code for converting db to voltage and the final noise figure and the IIP3:

```
db-v for NF:
NF1 (3.0 dB): 0.8934 V
NF2 (18.0 dB): 5.0238 V
NF3 (25.0 dB): 11.2468 V
NF4 (30.0 dB): 20.0000 V
NF5 (35.0 dB): 35.5656 V

db-v for gain:
gain1 (17.0 dB): 4.4774 V
gain2 (12.0 dB): 2.5179 V
gain3 (6.0 dB): 1.2619 V
gain4 (0.0 dB): 0.6325 V
gain5 (60.0 dB): 632.4555 V

db-v for IIP:
IIP1 (-10.0 dBm): 0.2000 V
IIP2 (0.0 dBm): 0.6325 V
IIP3 (12.0 dBm): 2.5179 V
IIP4 (20.0 dBm): 6.3246 V
IIP5 (25.0 dBm): 11.2468 V

==================== RESULTS =====================
System Noise Figure = 6.94 dB
System IIP3 = -21.81 dBm
==================================================
>>
```

Noise Figure:6.94 db

System IIP3:-21.81 db